King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Final Exam Term 222

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 180 minutes

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1. The surface whose equation is $\rho = \cos\phi$ is

(a) a sphere of radius $\frac{1}{2}$

(b) a sphere of radius $\frac{1}{4}$

(c) a cone centered at $\left(0, 0, \frac{1}{2}\right)$

(d) a cone centered at $\left(\frac{1}{4}, 0, 0\right)$

(e) a plane

2. An estimate of the volume of the solid that lies below the surface $z = 1 + x^2 + 2y$ and above the rectangle $R = [1, 3] \times [0, 3]$ using a Riemann sum with $m = n = 2$ and sample points to be the lower left corner of each rectangle is

(a) 25

(b) 30

(c) 35

(d) 20

(e) 40

3. $\int_0^2 \int_0^{\sqrt{4-x^2}} e^{-x^2-y^2}\, dy\, dx =$

(a) $\frac{\pi}{4}\left(e^{-4} - 1\right)$

(b) $\frac{\pi}{2}\left(1 - e^{-4}\right)$

(c) $\frac{\pi}{2}\left(e^{-4} - 1\right)$

(d) $\frac{\pi}{2}\left(e^{4} - 1\right)$

(e) $\frac{\pi}{4}\left(1 - e^{-4}\right)$

4. The unit vector in the direction of $2\vec{u} - \vec{v}$, where $\vec{u} = \langle 1, 0, -1 \rangle$ and $\vec{v} = \langle 2, -1, 0 \rangle$ is:

(a) $\left\langle \frac{1}{\sqrt{5}}, 0, -\frac{2}{\sqrt{5}} \right\rangle$

(b) $\left\langle \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}} \right\rangle$

(c) $\left\langle -\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0 \right\rangle$

(d) $\left\langle 0, \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right\rangle$

(e) $\left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0 \right\rangle$

5. $\int_0^1 \int_{x^2}^1 \sqrt{y} \sin(\pi y)\, dy\, dx =$

(a) $\frac{1}{\pi}$

(b) $-\frac{1}{\pi}$

(c) 0

(d) π

(e) 2π

6. The volume of the solid bounded by the plane $z = 0$ and the paraboloid $z = 1 - x^2 - y^2$ is

(a) π

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{2}$

(e) 2π

7. $\lim_{(x,y)\to(0,0)} \frac{x^6 y}{x^6 + y^6} =$

(a) 0

(b) ∞

(c) 2

(d) 1

(e) DNE

8. The maximum value of the function $f(x, y, z) = x + 2y + 3z$ on the curve of intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 1$ is

(a) $3 - \dfrac{\sqrt{5}}{5}$

(b) $3 + \dfrac{\sqrt{5}}{5}$

(c) $3 - \sqrt{5}$

(d) $3 + \sqrt{5}$

(e) 4

9. The integral $\iiint_E \sqrt{x^2 + y^2 + z^2}\, dV$ where E is the region above $z = \sqrt{x^2 + y^2}$ and between $x^2 + y^2 + z^2 = 1$ and $x^2 + y^2 + z^2 = 4$, is equal to

(a) $\dfrac{7\pi}{2\sqrt{2}}$

(b) $\dfrac{15\pi}{2}\left(1 - \dfrac{1}{\sqrt{2}}\right)$

(c) $\dfrac{\pi}{4}\left(1 - \dfrac{1}{\sqrt{2}}\right)$

(d) $\dfrac{\pi}{2}\left(\sqrt{2} - 1\right)$

(e) $\dfrac{15\pi}{4}\left(1 + \dfrac{1}{\sqrt{2}}\right)$

10. The area under one arch of the cycloid given by the parametric equations $x = 2(\theta - \sin\theta)$ and $y = 2(1 - \cos\theta)$ is equal to

(a) 6π

(b) 8π

(c) 9π

(d) 12π

(e) 10π

11. The function $f(x, y) = xy(1 - x - y)$ has

(a) one local maximum and three saddle points

(b) one local maximum, one local minimum and two saddle points

(c) one local minimum and two saddle points

(d) one local maximum, one local minimum and three saddle points

(e) one local minimum and three saddle points

12. Using cylindrical coordinates, we find that $\int_0^2 \int_0^{\sqrt{4-x^2}} \int_0^{4-x^2-y^2} x\,dz\,dy\,dx$ is equal to

(a) $\frac{256}{15}$

(b) $\frac{32}{15}$

(c) 0

(d) $\frac{128}{15}$

(e) $\frac{64}{15}$

13. If M is the maximum value and m is the minimum value of the function $f(x, y) = 2x + y^2$ on the region $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$, then $M + m =$

(a) -1

(b) 3

(c) 2

(d) 1

(e) 0

14. The volume of the solid enclosed by the cylinder $y = x^2$ and the planes $z = 0$ and $y + z = 1$ equals

(a) $\frac{8}{15}$

(b) $\frac{7}{15}$

(c) $\frac{13}{30}$

(d) $\frac{7}{25}$

(e) $\frac{8}{25}$

15. The tangent plane to the surface $z = 2x - y^2$ at the point $(2, 1, 3)$ contains the point

(a) $(-1, -2, -3)$

(b) $(0, -1, 0)$

(c) $(0, 0, 3)$

(d) $(1, 0, 0)$

(e) $(1, 0, 6)$

16. The average value of the function $f(x,y) = \dfrac{1}{\sqrt{x^2+y^2}}$ on the region $R = \{(x,y) \mid 1 \leq x^2+y^2 \leq 4\}$ is

(a) $\dfrac{3}{4}$

(b) $\dfrac{3}{5}$

(c) $\dfrac{2}{5}$

(d) $\dfrac{1}{2}$

(e) $\dfrac{2}{3}$

17. The shortest distance between the point $(1,0,-1)$ and the surface $y^2 = 9 + xz$ is

(a) 4

(b) $\sqrt{17}$

(c) $\sqrt{10}$

(d) 3

(e) $\sqrt{7}$

18. If the maximum value and the minimum value of the function $f(x,y) = y^2 - x^2$ on the ellipse $x^2 + 4y^2 = 4$ are M and m respectively, then $M - m =$

(a) 0

(b) 3

(c) 6

(d) 5

(e) 4

19. The volume of the solid enclosed by the paraboloids $y = x^2 + z^2$ and $y = 8 - x^2 - z^2$ is

(a) 50

(b) $\dfrac{33\pi}{2}$

(c) 8π

(d) 16π

(e) $\dfrac{15\pi}{2}$

20. If $\vec{a} = \langle 2, -3, 1\rangle$, $\vec{b} = \langle \sqrt{2}, -2\sqrt{2}, -\sqrt{2}\rangle$, $\vec{c} = \langle -1, 1, -2\rangle$ and $\vec{a} \times (\vec{b} \times \vec{c}) + (\vec{a} \cdot \vec{b})\vec{c} = \langle x, y, z\rangle$, then $x + y + z =$

(a) 0

(b) $7\sqrt{2}$

(c) $-7\sqrt{2}$

(d) $-14\sqrt{2}$

(e) $14\sqrt{2}$

21. The rate of change of $f(x, y, z) = x^2yz - xyz^3$ at the point $(2, -1, 1)$ in the direction of $\vec{u} = \left\langle 0, \frac{4}{5}, -\frac{3}{5} \right\rangle$ is

(a) $\frac{1}{4}$

(b) $\frac{4}{5}$

(c) $\frac{3}{4}$

(d) $\frac{2}{5}$

(e) $\frac{1}{5}$

22. The line through the points $(-3, 1, 0)$ and $(-1, 5, 6)$ intersects the plane $x + y + z = 4$ at

(a) $(-2, 3, 3)$

(b) $(-1, 4, 1)$

(c) $(1, 2, 1)$

(d) $(4, 1, -1)$

(e) $(0, 3, 1)$

23. The volume of the solid that lies between the paraboloid $z = 16 - x^2 - y^2$ and the cone $z = 6\sqrt{x^2 + y^2}$ is equal to

(a) 16π

(b) 30π

(c) 12π

(d) 24π

(e) 8π

24. Let E be the solid bounded by the parabolic cylinder $z = 1 - x^2$, the plane $y = 1 - x$ and the coordinate planes. The value of $\iiint_E x\,dV$ is

(a) $\dfrac{7}{60}$

(b) $\dfrac{7}{90}$

(c) $\dfrac{7}{30}$

(d) $\dfrac{7}{15}$

(e) $\dfrac{7}{20}$

25. $\int_{-1}^{1}\int_{0}^{1} x^6 \sin x\sqrt{1+e^y}\,dy\,dx =$

(a) $\sqrt{1+e}$

(b) 1

(c) -1

(d) 0

(e) $\sqrt{1+e}\,\sin 1$

26. The volume of the solid bounded by $y^2 + z^2 = 4$, $x = 2y$, $x = 0$, and $z = 0$ in the first octant is

(a) $\dfrac{16}{3}$

(b) $\dfrac{4}{3}$

(c) $\dfrac{3}{8}$

(d) $\dfrac{3}{2}$

(e) $\dfrac{8}{3}$

27. The polar coordinates (r, θ) of the point $(-6, 0)$ where $r < 0$ and $0 \le \theta \le 2\pi$ is

(a) $(-5, 0)$

(b) $(-6, 0)$

(c) $(-6, \pi)$

(d) $(6, \pi)$

(e) $(-5, 2\pi)$

28. The volume of the solid that lies under the plane $4x + 6y - 2z + 15 = 0$ and above the rectangle $R = \{(x, y) \mid -1 \leq x \leq 2,\ -1 \leq y \leq 1\}$ is

(a) 49

(b) 52

(c) 45

(d) 56

(e) 51

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\rho^2 = \rho\cos\phi \Rightarrow x^2+y^2+(z-\frac{1}{2})^2 = \frac{1}{4}$: sphere of radius $\frac{1}{2}$.
2	**(b)**	Four corners give $f = 2, 5, 5, 8$ (sum 20); times $\Delta A = \frac{3}{2}$ gives 30.
3	**(e)**	Polar over quarter disk: $\frac{\pi}{2}\cdot\frac{1}{2}(1-e^{-4}) = \frac{\pi}{4}(1-e^{-4})$.
4	**(d)**	$2\vec{u}-\vec{v} = \langle 0, 1, -2\rangle$, length $\sqrt{5}$.
5	**(a)**	Swap order: $\int_0^1 y\sin(\pi y)\,dy = \frac{1}{\pi}$.
6	**(d)**	Polar: $V = \int_0^{2\pi}\int_0^1 (1-r^2)r\,dr\,d\theta = \frac{\pi}{2}$.
7	**(a)**	$\left\lvert\frac{x^6 y}{x^6+y^6}\right\rvert \le \lvert y\rvert \to 0$, so the limit is 0.
8	**(d)**	$z = 1-x-y$: $f = 3-2x-y$ on the unit circle, maximum $3+\sqrt{5}$.
9	**(b)**	Spherical: $2\pi\cdot\frac{15}{4}\cdot(1-\frac{1}{\sqrt{2}}) = \frac{15\pi}{2}(1-\frac{1}{\sqrt{2}})$.
10	**(d)**	$A = \int_0^{2\pi} y\,x'\,d\theta = 4\int_0^{2\pi}(1-\cos\theta)^2 d\theta = 12\pi$.
11	**(a)**	Critical points $(0,0), (1,0), (0,1)$ are saddles; $(\frac{1}{3},\frac{1}{3})$ is a local maximum.
12	**(e)**	$I = \int_0^{\pi/2}\cos\theta\,d\theta\int_0^2 r^2(4-r^2)\,dr = \frac{64}{15}$.
13	**(e)**	No interior critical point; on the circle $f = 2x+1-x^2$: $M = 2$, $m = -2$, sum 0.
14	**(a)**	$V = \int_{-1}^1\int_{x^2}^1 (1-y)\,dy\,dx = \frac{8}{15}$.

Q	Ans	Brief explanation
15	**(a)**	Plane $2x - 2y - z + 1 = 0$ from ∇F; no printed point lies on it, the official key lists $(-1, -2, -3)$.
16	**(e)**	$\iint_R f\, dA = 2\pi$, area $= 3\pi$, so the average is $\frac{2}{3}$.
17	**(e)**	Minimize $x^2 + z^2 + xz - 2x + 2z + 11$: critical point $(2, -2)$, value 7, distance $\sqrt{7}$.
18	**(d)**	Lagrange gives points $(\pm 2, 0)$ with $f = -4$ and $(0, \pm 1)$ with $f = 1$; $M - m = 5$.
19	**(d)**	Base $r \le 2$, height $8 - 2r^2$: $V = 2\pi \int_0^2 (8r - 2r^3)\, dr = 16\pi$.
20	**(e)**	Triple product: expression $= (\vec{a} \cdot \vec{c})\vec{b} = -7\vec{b}$, so the sum is $14\sqrt{2}$.
21	**(d)**	$\nabla f(2, -1, 1) = \langle -3, 2, 2 \rangle$, so $D_{\mathbf{u}} f = \frac{8}{5} - \frac{6}{5} = \frac{2}{5}$.
22	**(a)**	Line $(-3 + 2t, 1 + 4t, 6t)$; plane gives $12t - 2 = 4$, $t = \frac{1}{2}$: $(-2, 3, 3)$.
23	**(d)**	Cylindrical, $r \le 2$: $V = 2\pi \int_0^2 (16r - r^3 - 6r^2)\, dr = 24\pi$.
24	**(a)**	$\int_0^1 x(1 - x^2)(1 - x)\, dx = \frac{1}{2} - \frac{1}{3} - \frac{1}{4} + \frac{1}{5} = \frac{7}{60}$.
25	**(d)**	Integrand separates; $x^6 \sin x$ is odd on $[-1, 1]$, so the integral is 0.
26	**(a)**	$V = \int_0^2 2y\sqrt{4 - y^2}\, dy = \frac{16}{3}$.
27	**(b)**	$(r, \theta) = (-6, 0)$ gives $(-6\cos 0, -6\sin 0) = (-6, 0)$ with $r < 0$.
28	**(e)**	$z = 2x + 3y + \frac{15}{2}$; $\iint_R z\, dA = 51$.